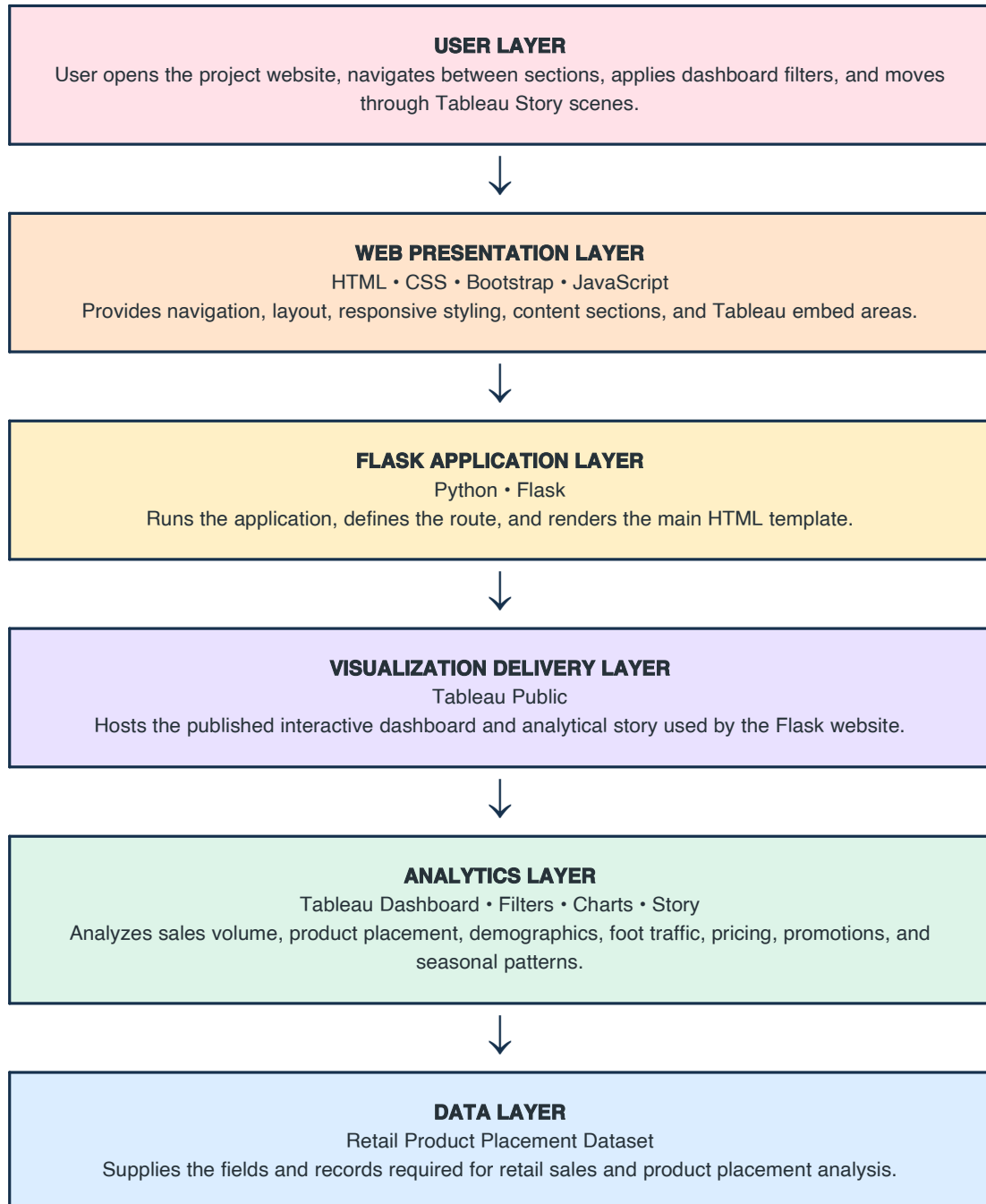


SOLUTION ARCHITECTURE

Strategic Product Placement Analysis

Architecture Overview

The solution architecture defines how the retail dataset, Tableau analytics, published visualizations, Flask application, frontend interface, and user interaction work together. The architecture separates the analytics layer from the web presentation layer while connecting them through embedded Tableau views.



ARCHITECTURE COMPONENTS

Responsibilities and Connections

Component	Responsibility	Connection
Retail Dataset	Stores product placement and sales-related fields used in the analysis.	Connected to Tableau for visualization and analysis.
Tableau Analytics	Creates charts, comparisons, filters, dashboard, and story.	Uses the dataset and produces interactive visual views.
Tableau Public	Publishes and hosts the dashboard and story.	Provides the published views embedded in the web interface.
Flask Application	Serves the project website and renders the HTML template.	Connects the browser request to the project interface.
Frontend Interface	Displays project sections, navigation, and Tableau embed containers.	Presents the Flask page and embedded Tableau content to the user.
User	Explores the dashboard, applies filters, and navigates story scenes.	Interacts with the web interface and Tableau visualizations.

End-to-End Architecture Flow

Retail Dataset → Tableau Analysis → Dashboard and Story → Tableau Public → Flask Web Application → User Interaction → Retail Insights

The retail dataset is analyzed in Tableau to create interactive visualizations. The dashboard and story are published to Tableau Public and embedded in the Flask website. The user accesses the web interface, explores the visualizations, and interprets sales and product placement patterns.

Architecture Outcome

The solution architecture provides a simple separation between data analysis and web presentation. Tableau handles the retail analytics and interactive visualizations, Tableau Public delivers the published views, Flask serves the application, and Bootstrap-based frontend technologies provide the user interface. This architecture supports a clear and interactive Strategic Product Placement Analysis experience.

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